Pure alternating kernels in the critical prefixes

Human Compression project: L044 working continuation

We retain the full critical six-point-head hypotheses, actual ambient IP and smaller-degree induction. Thus $n = 2r \geq 18$, the first head is supported on $D = \{1, \dots, 6\}$, and

$$\tau = \rho_3 = (6\ b)\delta, \quad \sigma = \rho_4, \quad R = G_{\geq 3} = \mathrm{Alt}(Z), \qquad Z = (\Omega \setminus D) \cup \{6\}.$$

The head is transitive on D, σ fixes D, and $G_{\geq 5}$ fixes $D \cup \{b\}$. The complete critical-tail proof gives $x = b^\sigma \neq b$ and $\tau x = x$ without assuming $G_{\geq 4}$ transitive. The involution δ is odd and fixes $D \cup \{b\}$. Every alternating group below is the actual natural subgroup fixing all points outside its displayed support.

Lemma 1 (A support-merging extension)**.** *Let P preserve C, contain $\mathrm{Alt}(C)$, and have $|C| \geq 4$. Let t be an involution with $|C \cap C^t| \geq 2$. Then $\langle P, t\rangle$ contains $\mathrm{Alt}(C^{\langle P,t\rangle})$, where the superscript denotes the union of all translates of C.*

Proof. Natural alternating groups on two supports of size at least four whose intersection contains two points generate the alternating group on their union. Indeed, choose two common points a, b. The three-cycles $(a\ b\ z)$ for every other point in the union belong to the generated group; these generate its full alternating group. The latter fact follows by conjugating such cycles to obtain all three-cycles, which generate every even permutation.

Starting with C, first adjoin C^t using this observation. Maintain a set U containing $C \cup C^t$ with $\mathrm{Alt}(U)$ in $\langle P, t\rangle$. For $p \in P$, both U and U^p contain C; for t, both U and U^t contain $C \cup C^t$. Thus we may adjoin the translate by any generator of P or by t and still have the full alternating group on the enlarged set. Finite iteration terminates at an invariant set, exactly the union of all translates of C. This proves the assertion without a primitive-group or Jordan theorem. □

Theorem 2 (Actual prefix kernels and their exterior groups)**.** *Put $P_k = G_{\leq k}$ and let C_k be its actual orbit containing D. Then*

$$C_3 = D \cup \{b\}, \qquad C_4 = D \cup \{b, x\}, \qquad \mathrm{Alt}(C_k) \leq P_k \quad (3 \leq k \leq r-1).$$

Let J_k be the literal permutation image of P_k on $\Omega \setminus C_k$. With restrictions to these disjoint sets,

$$P_k = \{(s, j) \in \mathrm{Sym}(C_k) \times J_k : \operatorname{sgn} s = \operatorname{sgn} j\}. \tag{1}$$

For $k \geq 4$ every generator of index at least $k + 2$ fixes C_k pointwise. The sets $C_3, C_4, \dots, C_{r-1}$ strictly increase, and $C_{r-1} = \Omega$.

Proof. Conjugating the single core transposition $(6\ b)$ by the transitive head on D gives every $(d\ b)$ in the core image. Products of two conjugates of τ cancel the exterior δ and give the pure three-cycles on $D \cup \{b\}$. Thus P_3 preserves this seven-point set, is transitive there, and contains its full alternating group. The pure-overlap/IP argument gives $\tau x = x$, so every generator of P_3 fixes x. The permutation σ fixes D and exchanges b, x. Consequently P_4 preserves $D \cup \{b, x\}$ and contains its alternating group, by merging the conjugate seven-point supports. This proves both initial orbit identities.

Suppose $k \geq 5$ and the conclusion is proved for $k-1$ and $k-2$. String commutation makes ρ_k centralize $\mathrm{Alt}(C_{k-2})$, and hence fix that set pointwise: the centralizer of natural $\mathrm{Alt}(C)$ on its support is trivial when $|C| \geq 4$. Here $|C_{k-2}| \geq 7$, so C_{k-1} and its ρ_k-translate intersect in at least seven points. Lemma 1, applied to P_{k-1}, gives $\mathrm{Alt}(C_k) \leq P_k$. Its invariant union is exactly the orbit C_k, since P_{k-1} is transitive on C_{k-1}. The same centralizer argument proves the stated fixation by every later distant generator.

For (1), all elements of P_k are even, so the two restriction signs agree. Given any $j \in J_k$, choose one lift $(s, j) \in P_k$. Multiplication by the full kernel $\mathrm{Alt}(C_k) \times 1$ supplies exactly all permutations on C_k with sign equal to that of j. This proves equality with the displayed set, including the case that every element of J_k is even.

If $C_k = C_{k-1}$ for some $k \geq 4$, then P_k preserves C_{k-1} and all generators of index at least $k+1$ fix it, because they centralize $\mathrm{Alt}(C_{k-1})$. Transitivity of G would force $C_{k-1} = \Omega$. But then P_{k-1} contains $\mathrm{Alt}(\Omega)$, equals G, and contradicts independence of the actual string C-marking. Thus equality cannot occur. Finally $C_{r-1} = \Omega$ by transitivity, proving all assertions. □

Lemma 3 (The exact local omission bridge)**.** *For $j \geq 3$, put $R_j = \langle \rho_k : k \geq 3,\ k \neq j \rangle$ and let O_6 be its orbit of 6 on Z. The G_j-orbits are $O_6 \cup (D \setminus \{6\})$ and all other R_j-orbits on Z. Consequently every j-edge has the same eligibility for its ambient omission G_j and its actual local omission R_j. Moreover, for $k \geq 3$,*

$$G_{\{3,\ldots,k\}} = P_k \cap R = \{(s, j) \in \mathrm{Sym}(C_k \setminus (D \setminus \{6\})) \times J_k : \operatorname{sgn} s = \operatorname{sgn} j\}. \tag{2}$$

Proof. We have $G_j = \langle G_{<3}, R_j \rangle$. The head moves only D, meets Z at the single vertex 6, and is transitive on D. It therefore adjoins the other five head vertices to O_6 and merges no other R_j-orbits. Since ρ_j is supported on Z, eligibility is unchanged. Formula (2) is (1) restricted to permutations fixing $D \setminus \{6\}$ pointwise; R is exactly that natural alternating group, and actual ambient IP supplies the marked intersection. No quotient intersection property is asserted. □

Attribution and remaining scope. Root proposed the initial eight-point parity-fiber description and suggested extending the pure support. The finite support-merging induction and complete orbit/fiber formulas above implement that suggestion; a distinct review must disclose this contribution. All complete critical-head and pure-overlap proofs remain paid. At a later unshared cut with transitive supported prefix, (1) now identifies that prefix as natural alternating directly, because its exterior image is trivial. A shared cut may retain a nontrivial J_k; it is not silently deleted. The index-3 prefix and the remaining opposite-end and source rigidity obligations are not resolved by these formulas. No new computation, formalization, whole-original credit or question claim is made.